

17 — Comprehensive Mentor Presentation Script

■ Ready-to-Speak Script for Project Defense & Oral Evaluation

This document contains a structured presentation script written specifically for a student presenting the **Maritime Port Intelligence Platform (MPIP)** to an academic or technical mentor.

1. Opening

*"Good morning/afternoon, Respected Mentor. Today I am presenting our project: the **Maritime Port Intelligence Platform (MPIP)**.*

At a high level, this is an AI-powered, data-grounded decision support system designed for maritime port authorities, terminal operators, and customs agencies. It transforms historical port traffic records into forward-looking predictive insights."

2. Problem Statement

"What problem are we solving? In traditional port operations, management relies heavily on fragmented spreadsheets and delayed monthly PDF reports. Port executives can see which vessels are docked at anchor today, but they lack predictive visibility into how much cargo tonnage of specific commodities will arrive next month or next quarter. This leads to severe berth congestion, vessel queue delays, misallocated crane crews, and supply chain disruptions for hinterland industries that depend on port raw materials."

3. Proposed Solution

"How does our system solve this? We built a full-stack decision-support system featuring a FastAPI microservice backend and a Streamlit interactive frontend. The system ingests multi-year historical monthly traffic datasets, applies time-series machine learning models, and outputs predictive cargo volume forecasts, commodity demand trends, risk alerts, and interactive What-If scenario simulations."

4. Cargo Focus

*"How is cargo central to our project? Our project's primary axis is **Predictive Cargo Analytics**. Rather than just counting generic ship arrivals, every chart, forecast, and calculation is measured in **Cargo Tonnes** broken down*

across 16+ specific commodities—such as Crude Oil, Thermal Coal, Iron Ore, Containers, and Fertilizers. Furthermore, every commodity is mapped directly to actual port terminal infrastructure, berth draft depths, and hinterland industrial consumers like the MRPL Refinery and UPCL Power Plant."

5. Technical Architecture

*"How does the system work technically?"

* **Backend Layer:** Built using **FastAPI** running on Uvicorn. It organizes business logic across 12 feature routers (**CODE_PH_0**, **CODE_PH_1**, **CODE_PH_2**, **CODE_PH_3**, **CODE_PH_4**, etc.).

* **Frontend Layer:** Built using **Streamlit** and **Plotly**. It features a custom Warm Amber / Cream design system with 9 horizontal navigation tabs.

* **API Integration:** The frontend communicates asynchronously with backend REST endpoints via HTTP GET and POST requests. "*"

6. Data Source & Processing

"Where does the data come from? Our platform is grounded in real historical monthly cargo records from **New Mangalore Port Authority (NMPA)** (**CODE_PH_0**), containing 1,354 records spanning 2021, 2023, 2024, 2025, and 2026. The backend pandas pipeline handles missing values, cleans raw tonnage figures, and constructs continuous month-by-month time series."

7. Analytics & Calculations

"What analysis are we performing? We perform commodity-wise traffic aggregation, Year-over-Year (YoY) growth calculations, 3-month Exponential Moving Average (EMA) trend signal classification (**CODE_PH_0** / **CODE_PH_1** / **CODE_PH_2**), trade lane velocity tracking, and berth utilization modeling."

8. AI/ML Implementation

*"Where is AI/ML being used?"

1. **Forecasting Model:** We built a dual time-series model combining Holt-Winters Exponential Smoothing (for seasonality) and Multi-Feature Ridge Regression with L2 regularization. It projects volume up to 12 months ahead with 95% dynamic confidence interval bands.

2. **Model Validation:** We perform chronological out-of-sample backtesting against a Seasonal Naive baseline model, measuring Weighted Absolute Percentage Error (**WAPE**), MAPE, MAE, and RMSE.

3. **Explainability:** The system quantifies forecast driver weights—showing exactly how much monsoon swells, hinterland demand, or trade velocity contribute to a forecast.

4. **AI Copilot:** We implemented a 3-tier cognitive query dispatcher supporting natural language interaction across 5 specialized agent roles."

9. System Improvements vs Previous Version

*"What did we change from the previous version of the project?"

* Previously, our project was a generic vessel ETA counter with static mock location dots.

* In the current version, we made **Predictive Cargo Analytics** our clearly visible core capability.

* We replaced simple linear estimates with a mathematically backtested Holt-Winters + Ridge ML pipeline.

* We added an interactive What-If Cargo Scenario Simulator and aligned supporting tabs like Trade Intelligence and Anomaly Detection to explicitly focus on commodity demand and cargo risk."

10. Key Benefits Provided

*"What improvements does this new version provide?"

1. **Proactive Planning:** Port authorities gain forward visibility 3, 6, and 12 months ahead.

2. **Reduced Demurrage:** Terminal operators can pre-allocate berths and cranes based on projected tonnage.

3. **Interactive Contingency Testing:** Managers can simulate vessel arrival spikes or weather delays before disruptions occur."

11. Future Scope

"What can be added next? In future iterations, we can integrate live IoT sensor feeds from crane gantries, implement PostgreSQL/TimescaleDB database persistence, and deploy real-time AIS satellite API streams."